

Implement a FPTAS for the 0-1 Knapsack Problem

Implement the classic FPTAS for 0-1 Knapsack using value scaling.

- Input: List of items with weights and values, knapsack capacity, and a precision parameter ϵ .
- Output: Total value of an approximate solution within $(1-\epsilon) \times \text{OPT}$.

Tasks:

- Scale item values using ϵ .
- Use dynamic programming on scaled values.
- Translate back to get approximate solution.

Sample Input	Sample Output
3 50 0.5 10 60 20 100 30 120	220 1 2
Explanation: First line: number of items, knapsack capacity, epsilon. Next 3 lines: weight value pairs.	Explanation: First line is total cost, second line is the tour.